Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the pat of the question) from the top of a new page.
- Write legibly and in orderly fashion maintaining all mathematical norms and rules. Prepare a single solution file.
- Start working right away. There is no late submission form. If you miss the deadline, you need to use the make-up assignment to cover up the marks.

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1. In the classes, we discussed three forms of floating number representations as shown below,

$$\text{Standard Form} : \quad F = \pm(0.d_1 d_2 d_3 \cdots d_m)_\beta \, \beta^e \, , \tag{1}$$
$$\text{IEEE Denormalized Form} : \quad F = \pm(1.d_1 d_2 d_3 \cdots d_m)_\beta \, \beta^e \, , \tag{2}$$
$$\text{IEEE Normalized Form} : \quad F = \pm(0.1 d_1 d_2 d_3 \cdots d_m)_\beta \, \beta^e \, , , \tag{3}$$

where $d_i, \beta, e \in \mathbb{Z}$, $0 \leq d_i \leq \beta - 1$ and $e_{\min} \leq e \leq e_{\max}$. Now, let's take, $\beta = 2$, $m = 4$ and $-4 \leq e \leq 2$. Based on these, answer the following:

(a) (3 marks) What are the maximum numbers that can be stored in the system by the three forms defined above?

(b) (3 marks) What are the non-negative minimum numbers that can be stored in the system by the three forms defined above?

(c) (4 marks) Using Eq.(1), find all the decimal numbers for $e = -3$, plot them on a real line, and show if the number line is equally spaced or not.

2. Let $\beta = 2, m = 3, e_{\min} = -2$ and $e_{\max} = 2$. Answer the following questions:

(a) (4 marks) Find the floating point representation of the numbers $(2.23)_{10}$ and $(2.2018)_{10}$ in the Denormalized form.

(b) (2 marks) Compute the rounding errors for Part (a).

(c) (4 marks) Can the numbers $(2.23)_{10}$ and $(2.2018)_{10}$ be represented in normalized form? If so, find the floating-point representations. If not, then concisely explain why?

Motto: Mathematics is NOT difficult, but what is difficult is to believe that mathematics is NOT difficult.